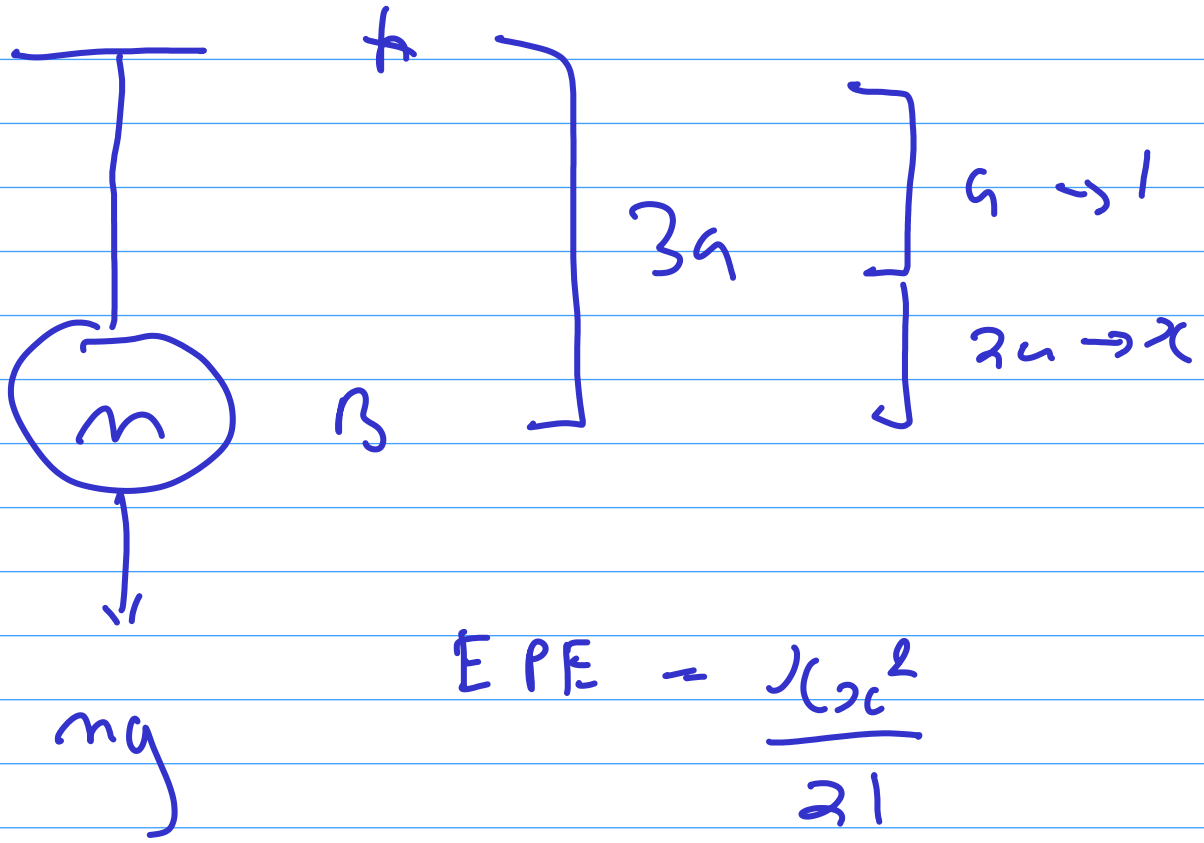
1.



$$EPE = \frac{\lambda (2a)^2}{2l}$$

At B → A

$K = 0$	$K = 0$
$W = 0$	$W = 0$
$E = \frac{\lambda (2a)^2}{2a}$	$E = \frac{\lambda (a)^2}{2a}$
$b = 0$	$G = mg \times 3a$

$$\frac{4\lambda a^2}{2a} = \frac{\lambda a^2}{2a} + mg\,3a$$

$2x\,a = \frac{1}{2}x\,a + 3mg\,a$

$2x = \frac{1}{2}x + 3mg$

$\frac{3}{2}x = 3mg$ (5/5)

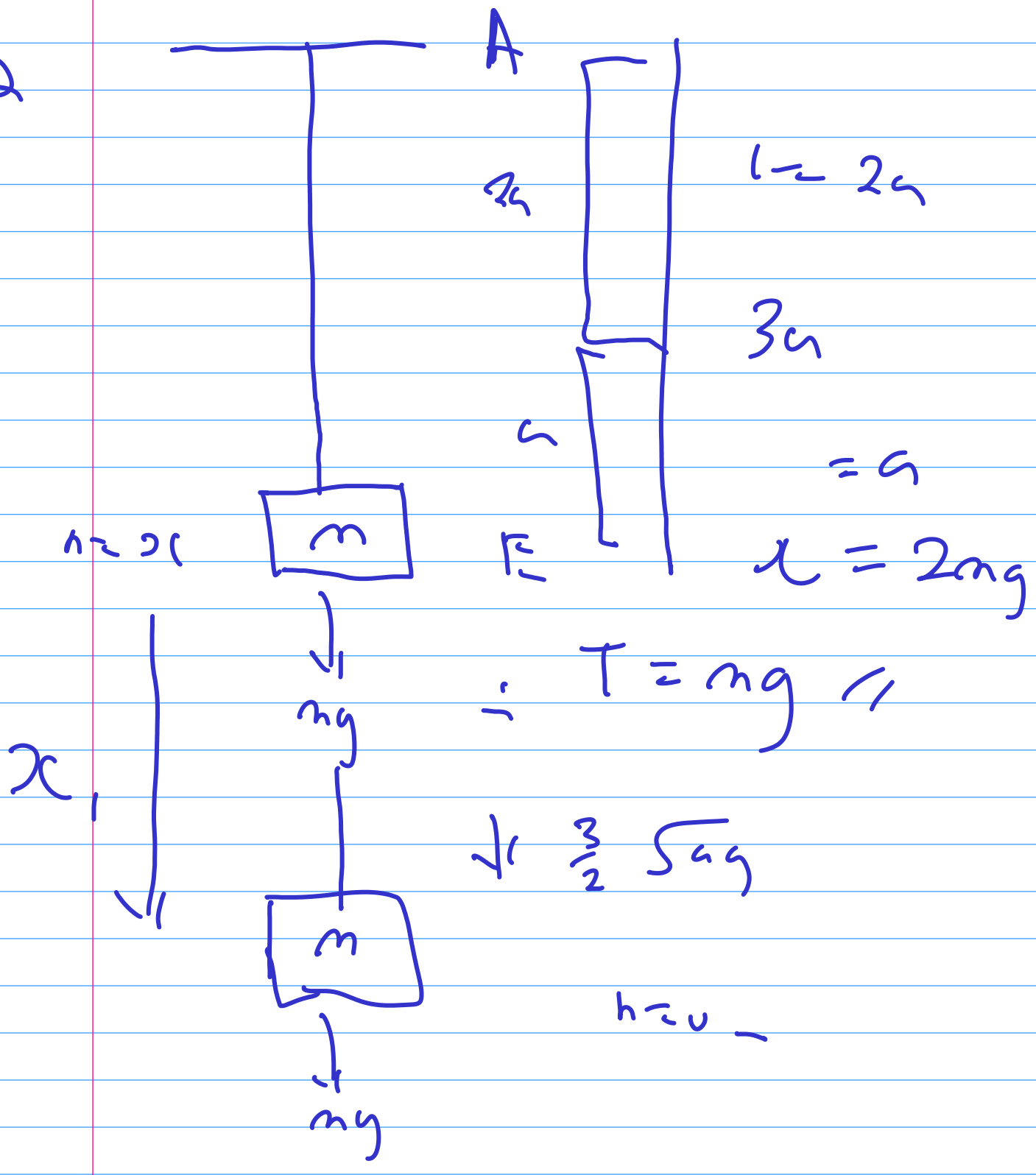
$x = 2mg$ N as required 1/1

b don't know 0/6

c it means that GPE would change as we would now have to model the spring 1/1

6/12

2



Extension =

$K = \frac{1}{2} m \left(\frac{3}{2} \sqrt{ag}\right)^2 \qquad K = 0$

$W = U \qquad W = 0$

$E = \frac{(2mg)(a)^2}{2(2a)} = \qquad E = \frac{(2mg)(a+x)^2}{2(2a)}$

$G = mg(x) \qquad G = 0$

$$\frac{9}{8} mag + \frac{2ma^2 g}{4a} + mgx = \frac{2mg(a+x)^2}{4a}$$

$$\frac{9}{8} mag + \frac{1}{2} mag + mgx = \frac{2mg(a^2 + 2ax + x^2)}{4a}$$

$$\frac{13}{8} mag + mgx = \frac{2mg(a^2 + 2ax + x^2)}{4a}$$

$\div mg$

$$\frac{13}{8} a + x = \frac{2(a^2 + 2ax + x^2)}{4a}$$

$$\frac{13}{8}a + 2x = \frac{2a^2 + 4ax + 2a^2}{4a}$$

$$\frac{13}{2}a^2 + 16ax = 2a^2 + 4ax + 2x^2$$

$$\frac{9}{2}a^2 = 2x^2$$

$$\frac{9}{4}a^2 = x^2$$

$$\frac{9}{4}a^2 - x^2 = 0$$

$$\left(\frac{3}{2}a - x\right)\left(\frac{3}{2}a + x\right) = 0$$

$$x = \frac{3}{2}a \quad \therefore$$

$$E_{PE} = \frac{2mg\left(a + \frac{3}{2}a\right)^2}{4a}$$

$$E_{PE} = \frac{2mg\left(\frac{5}{2}a\right)^2}{4a}$$

$$E_{PE} = \frac{2mg\left(\frac{25}{4}a^2\right)}{4a}$$

$$= \frac{\frac{25}{2}ma^2g}{4a}$$

$$= \frac{25}{8}mag \quad J$$

7/7

3